Context-Aware Voice Security Agent for Real-Time Credential Disclosure Detection

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1 Problem Statement

Voice-security systems can redact sensitive transcripts and detect scam-call patterns (Deepgram; Google). Can an agent also warn a user before they disclose a credential?

We investigate whether a voice agent can detect a user's intention to disclose credentials and interrupt before disclosure. Our English-language prototype focuses on passwords, persistent passcodes, and partial credentials. We compare rules, a lightweight classifier, and their combination on coverage, false warnings, and timing. One-time codes, speaker verification, and general scam classification are outside scope.

Shen et al. (2025) use LLM judgments after each turn to detect scams in Chinese conversations. We instead explore credential-specific warnings using English partial transcripts and a local harness. Our midpoint contribution is a working prototype, trained classifier, and three-way evaluation.

2 Methodology

Harness architecture. Deepgram Nova-3 streams partial transcripts. The harness checks these with visible dialogue history, using local rules and fine-tuned MiniLM. It records WAIT, WATCH, or INTERVENE, the source, and the reason. An intervention plays a preloaded warning and ends the simulated call. In the illustrative push-to-talk demo, Whiskey the Cat speaks to fictional caller Scam Ninja, whose responses use Mistral Small 3.2 through OpenRouter and Deepgram Aura-2 speech. Figure 1 illustrates the architecture.

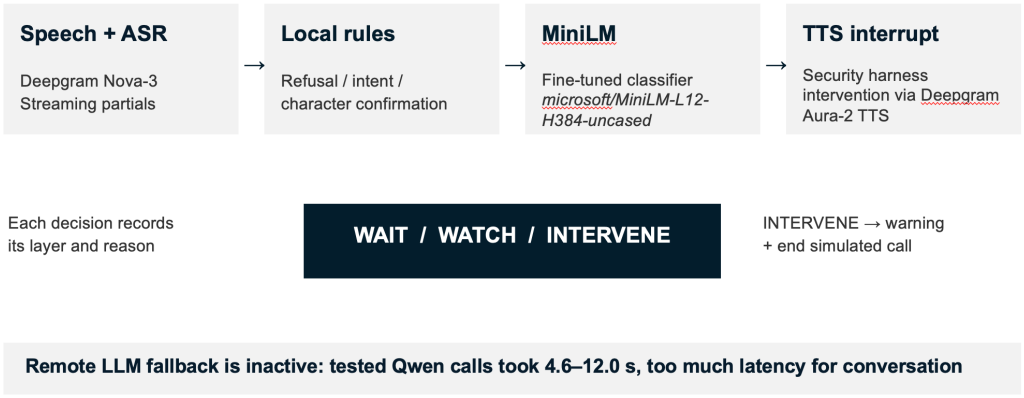


Figure 1: Security harness architecture. Rules can decide without invoking MiniLM; an intervention plays a preloaded warning rather than generating speech at decision time. The remote LLM fallback is inactive.

The harness checks rules first. If a rule returns a decision, MiniLM is skipped. Otherwise, MiniLM evaluates the conversation history and current partial transcript. This saves model

computation when a rule matches. However, if a rule returns WATCH for an incomplete phrase such as "It's…", MiniLM must wait for the next transcript update; this can delay a warning.

Data and training. We generated 1,000 synthetic English records in 500 safe/risky pairs. Source inspiration comprised 300 CallCenterEN families, 100 NCSU/FTC robocall families, 43 YouTube scam-baiting families, and 57 HarperValleyBank families (AIxBlock; Prasad and Reaves, 2023; rajkirant; Wu et al., 2020). Mistral received source types and coarse cue tags and generated synthetic dialogue, fictional credentials, labels, and text-event timestamps. Safe contrasts include refusals, reset assistance, private typing, tutorial examples, and non-login sharing. Approximately 20% of records received author spot checks; 40 audit-flagged risky records were repaired.

A fixed family split assigns 700/150/150 records to training/validation/test, keeping related safe/risky records together. The test set has 75 risky and 75 safe scenarios. Earlier work used 38 development scenarios to build replay and initial rules; current classifier training uses the repaired 1,000-record dataset.

We fine-tuned microsoft/MiniLM-L12-H384-uncased as a binary classifier over history and current text. Each risky training record supplies its annotated warning prefix; each safe record supplies its final prefix. Training therefore does not supervise every partial. Inputs exclude labels, provenance, and future events and use a 192-token limit. Three epochs were run; epoch 3 was selected by validation loss. Validation selected an upper threshold of 0.94983 to maximize recall subject to a 10% false-warning constraint. Runtime scores above that threshold produce INTERVENE, scores at or below 0.10 produce WAIT, and others produce WATCH. Scores have not been calibrated as deployment probabilities.

Approach that did not meet the latency goal. We initially connected Qwen3-14B through OpenRouter as an asynchronous fallback. Four early smoke-test calls took approximately 4.6–12.0 seconds each; in the risky example, the warning followed the complete password. This configuration was unsuitable for that example and is inactive in the live microphone flow. The current detector relies on local rules and MiniLM.

3 Results

Experimental setup. We compare three detection schemes: rules only, MiniLM only, and a rules-first combination of rules and MiniLM. Rules-only includes refusal, context, and fast-path rules; MiniLM-only bypasses them. Both model conditions use the same checkpoint and thresholds. For 150 synthetic test scenarios, Deepgram Aura-2 generated audio for both sides of the conversation, and Nova-3 streaming ASR transcribed the focal user turn; speaker roles were supplied. The comparison replays the same cached histories, ASR partials, and arrival times through all three schemes. Local computation is measured during replay and combined with saved arrival times using simulated serial queuing. This is an exploratory evaluation, not a new live end-to-end measurement or untouched final benchmark.

We report risk-warning coverage (risky scenarios receiving at least one warning divided by 75) and false-warning rate (safe scenarios receiving a warning divided by 75). A warning counts here whether or not its exact timing is appropriate.

Detection scheme	Risky warned	Risky missed	Safe falsely warned
Rules only	43/75 (57.3%)	32	2/75 (2.7%)
MiniLM only	74/75 (98.7%)	1	0/75 (0% observed)
Rules + MiniLM	75/75 (100%)	0	2/75 (2.7%)

Timing definition. T1 is the estimated time when the user begins saying the password, identified by locating the first password word in the transcript. It marks password onset, not the earliest moment when a warning could be justified. T3 is the estimated time when the system completes its warning decision. T3-T1 indicates how early or late that decision occurs relative to estimated password onset.

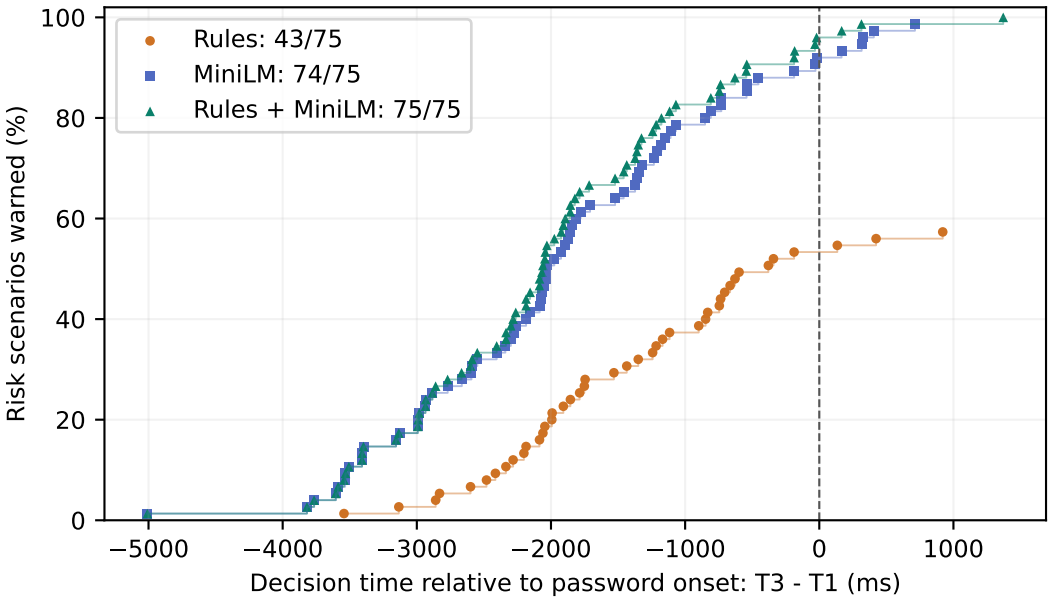


Figure 2: The x-axis shows T3 - T1 in milliseconds: smaller values indicate earlier decisions, and negative values indicate decisions before estimated password onset. The y-axis shows the cumulative percentage of all 75 risk cases warned by that time; higher means more risks were warned. Misses remain in the denominator. Ideally the curve is higher and further left, provided warnings are justified. The dashed line marks estimated password onset.

Findings. First, context-sensitive classification substantially increased coverage: rules missed 32 risky scenarios, compared with one for MiniLM. Second, the cascade achieved the highest observed coverage but inherited two safe-case false warnings; combining detectors did not improve every metric. Third, routing can delay warnings. In one case, the partial "It's" caused a rule to return WATCH, skipping MiniLM. MiniLM-only warned at approximately 323 ms after candidate password onset, whereas the cascade waited for another ASR partial and warned at 1,371 ms. The next partial arrived about 1.05 seconds later, while its MiniLM computation took approximately 6 ms. Faster classifier inference alone would not remove that waiting time.

Context is necessary to distinguish disclosure from benign discussion. Detecting a risk is only the first step: the warning also needs to reach the user early enough. This raises an accuracy–latency trade-off: waiting for more context may improve accuracy but delay intervention, while earlier decisions may increase false warnings. ASR uncertainty and cascade routing also affect performance.

Limitations. The data are synthetic and relatively structured, so real conversations may be messier. T3-T1 is approximate: it measures estimated decision time relative to estimated password onset, not when a user hears the warning; playback and audible output add delay. Both the rules and classifier have room for improvement and will be optimized for the final report.

4 Assets

Live prototype: https://call-guard-demo.vercel.app/ — interactive Scam Ninja / Whiskey the Cat voice demo.

Demo video: https://youtu.be/NzATwahUXQ0.

Code: https://github.com/cchenzhao/voice-security-agent — public repository with the detector, trained checkpoint, training scripts, fixed data/split, and comparison results. Eval-

uation and plotting code is under research/Proposal/security_harness/runs/timeliness_three_way/.

Reproduction: research/README.md documents commands to rerun the comparison from cached observations without paid API calls; local inference timing varies by machine.

5 Timeline

The remaining work is organized into milestones before final submission, with audio detection conditional on completing the timing evaluation.

Milestone	Work and completion criterion
Next week	Improve classifier and rules using observed refusal, indirect-answer, and character-confirmation failures.
Following week	Randomize synthetic examples and evaluation data; preserve family separation, freeze an independent evaluation set, and measure warning playback.
Final phase	Pilot a frozen audio encoder and trainable risk detector if feasible; compare coverage, false warnings, and timing, then finalize analysis and demo.

Latency approaches and their current status:

Approach	Current status and next adjustment
Streaming ASR	Active; examine partial-arrival gaps and revisions.
Local rules and MiniLM	Active; test model review during WATCH.
Preloaded warning audio	Active; measure browser and acoustic onset.
Remote LLM fallback	Inactive in the live flow after high-latency smoke tests.
Direct audio detection	Aspirational: frozen audio encoder with a trainable risk detector.

The audio extension would train a small detector over a frozen model's internal representations, rather than train a speech foundation model from scratch. Its benefit remains an empirical question: removing transcription may reduce one source of waiting but can introduce other accuracy and compute costs.

References

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AI-use disclosure. Codex assisted with implementation, training and evaluation scripts, debugging, experiment execution, and report drafting. Mistral generated synthetic dialogue and draft labels; AI also assisted repairs and structural checks. The author tested the prototype, identified failure cases, and spot-checked approximately 20% of the synthetic records.